TestAmerica

THE LEADER IN ENVIRONMENTAL TESTING

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ANALYTICAL REPORT

TestAmerica Laboratories, Inc.
TestAmerica Michigan
10448 Citation Drive
Suite 200
Brighton, MI 48116
Tel: (810)229-2763

TestAmerica Job ID: 190-17993-1
Client Project/Site: City of Cadillac WWTP

For:
City of Cadillac Utilities
1121 Plett Road
Cadillac, Michigan 49601

Attn: Cindy Tomaszewski

Patrick O'Meara

Authorized for release by:
12/10/2018 6:34:34 PM
Patrick O'Meara, Manager of Project Management
(330)966-5725
patrick.omeara@testamericainc.com

Designee for

Sue Schafer, Project Manager II
(810)229-2763
sue.schafer@testamericainc.com

This report has been electronically signed and authorized by the signatory. Electronic signature is intended to be the legally binding equivalent of a traditionally handwritten signature.

Results relate only to the items tested and the sample(s) as received by the laboratory.

Table of Contents

Sample Summary

Lab Sample ID	Client Sample ID	Matrix	Collected	Received
190-17993-1	Effluent	Water	11/05/18 13:15	11/06/18 09:35
190-17993-2	Effluent Duplicate	Water	11/05/18 13:15	11/06/18 09:35
190-17993-3	Field Blank	Water	11/05/18 13:10	11/06/18 09:35

Case Narrative

Job ID: 190-17993-1

Laboratory: TestAmerica Michigan

Narrative

Job Narrative
190-17993-1

Comments
No additional comments.

Receipt
The samples were received on 11/6/2018 9:35 AM; the samples arrived in good condition, properly preserved and, where required, on ice. The temperature of the cooler at receipt was 3.1° C.

No analytical or quality issues were noted, other than those described in the Definitions/Glossary page.

ANALYSIS REPORT

Prepared by:

Eurofins Lancaster Laboratories Environmental
2425 New Holland Pike
Lancaster, PA 17601

Prepared for:

TestAmerica
10448 Citation Drive
Suite 200
Brighton MI 48116

Report Date: November 29, 2018 16:22

Project: City of Cadillac WWTP

Account #: 01042
Group Number: 2010210
PO Number: 19001884
State of Sample Origin: MI

Electronic Copy To TestAmerica Attn: Sue Schafer

Respectfully Submitted,

Wendy A. Kozma

Wendy A. Kozma
Principal Specialist Group Leader

(717) 556-7257

To view our laboratory's current scopes of accreditation please go to http://www.eurofinsus.com/environment-testing/laboratories/eurofins-lancaster-laboratories-environmental/resources/certifications/ . Historical copies may be requested through your project manager.

SAMPLE INFORMATION

Client Sample Description	Sample Collection Date/Time	ELLE#
Effluent (190-17993-1) Water	11/05/2018 13:15	9903361
Effluent Duplicate (190-17993-2) Water	11/05/2018 13:15	9903362
Field Blank (190-17993-3) Water	11/05/2018 13:10	9903363

The specific methodologies used in obtaining the enclosed analytical results are indicated on the Laboratory Sample Analysis Record.

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Sample Description: **Effluent (190-17993-1) Water**
City of Cadillac WWTP

Project Name: **City of Cadillac WWTP**

TestAmerica
ELLE Sample #: **WW 9903361**
ELLE Group #: **2010210**
Matrix: Water

Submittal Date/Time: 11/16/2018 11:40
Collection Date/Time: 11/05/2018 13:15

CAT No.	Analysis Name	CAS Number	Result		Method Detection Limit	Dilution Factor
LC/MS/MS Miscellaneous	**EPA 537 Version 1.1 Modified**		**ng/l**		**ng/l**	
14473	4:2 fluorotelomersulfonate	757124-72-4	N.D.		0.84	1
14473	6:2 fluorotelomersulfonate	27619-97-2	1.6	J	0.84	1
14473	8:2 fluorotelomersulfonate	39108-34-4	N.D.		1.7	1
14473	NEtFOSAA	2991-50-6	N.D.		0.84	1
	NEtFOSAA is the acronym for N-ethyl perfluorooctanesulfonamidoacetic Acid.					
14473	NMeFOSAA	2355-31-9	1.8	J	0.84	1
	NMeFOSAA is the acronym for N-methyl perfluorooctanesulfonamidoacetic Acid.					
14473	Perfluorononanesulfonate(Pfns)	68259-12-1	N.D.		0.50	1
14473	Pfba-Perfluorobutanoic Acid	375-22-4	21		1.7	1
14473	Pfbs-Perfluorobutanesulfonate	375-73-5	58		0.25	1
14473	Pfda-Perfluorodecanoic Acid	335-76-2	1.3	J	0.76	1
14473	Pfdoda-Perfluorododecanoic	307-55-1	N.D.		0.42	1
14473	Pfds-Perfluorodecanesulfonate	335-77-3	N.D.		0.50	1
14473	Pfhpa-Perfluoroheptanoic Acid	375-85-9	6.6		0.34	1
14473	Pfhps-Perfluoroheptanesulfonat	375-92-8	N.D.		0.34	1
14473	Pfhxa-Perfluorohexanoic Acid	307-24-4	68		0.34	1
14473	Pfhxs-Perfluorohexanesulfonate	355-46-4	15		0.34	1
14473	Pfna-Perfluorononanoic Acid	375-95-1	1.2	J	0.34	1
14473	Pfoa-Perfluorooctanoic Acid	335-67-1	20		0.25	1
14473	Pfosa-Perfluorooctanesulfonami	754-91-6	N.D.		0.42	1
14473	Pfos-Perfluorooctanesulfonate	1763-23-1	6.5		0.34	1
14473	Pfpea-Perfluoropentanoic Acid	2706-90-3	22		1.7	1
14473	Pfpes-Perfluoropentanesulfonat	2706-91-4	8.2		0.34	1
14473	Pfteda-Perfluorotetradecanoic	376-06-7	N.D.		0.25	1
14473	Pftrda-Perfluorotridecanoic Ac	72629-94-8	N.D.		0.34	1
14473	Pfunda-Perfluoroundecanoic Aci	2058-94-8	N.D.		0.34	1

The recovery for labeled compound used as extraction standards is outside of QC acceptance limits in the method blank associated with this sample as noted on the QC Summary.

The recovery for the labeled compound used as extraction standards is outside the QC acceptance limits as noted on the QC Summary. The following corrective action was taken: The sample was reextracted outisde the required holding time and extraction standards were within QC acceptance limits. However, a sample injection standard was outside the QC acceptance limits in the re-extraction. The data is reported from the original extraction. Both sets of data are included in the data package.

Laboratory Sample Analysis Record

CAT No.	Analysis Name	Method	Trial#	Batch#	Analysis Date and Time	Analyst	Dilution Factor
14473	PFAS in Water by LC/MS/MS	EPA 537 Version 1.1 Modified	1	18321001	11/19/2018 11:30	Jason W Knight	1
14091	PFAS Water Prep	EPA 537 Version 1.1 Modified	1	18321001	11/17/2018 08:50	Courtney J Fatta	1

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Lancaster Laboratories Environmental

Analysis Report

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Sample Description: Effluent Duplicate (190-17993-2) Water
City of Cadillac WWTP

Project Name: City of Cadillac WWTP

TestAmerica
ELLE Sample #: WW 9903362
ELLE Group #: 2010210
Matrix: Water

Submittal Date/Time: 11/16/2018 11:40
Collection Date/Time: 11/05/2018 13:15

CAT No.	Analysis Name	CAS Number	Result	Method Detection Limit	Dilution Factor
LC/MS/MS Miscellaneous	**EPA 537 Version 1.1 Modified**		ng/l	ng/l	
14473	4:2 fluorotelomersulfonate	757124-72-4	N.D.	0.86	1
14473	6:2 fluorotelomersulfonate	27619-97-2	1.8	0.86	1
14473	8:2 fluorotelomersulfonate	39108-34-4	N.D.	1.7	1
14473	NEtFOSAA	2991-50-6	N.D.	0.86	1
	NEtFOSAA is the acronym for N-ethyl perfluorooctanesulfonamidoacetic Acid.				
14473	NMeFOSAA	2355-31-9	1.1 J	0.86	1
	NMeFOSAA is the acronym for N-methyl perfluorooctanesulfonamidoacetic Acid.				
14473	Perfluorononanesulfonate(Pfns)	68259-12-1	N.D.	0.52	1
14473	Pfba-Perfluorobutanoic Acid	375-22-4	20	1.7	1
14473	Pfbs-Perfluorobutanesulfonate	375-73-5	59	0.26	1
14473	Pfda-Perfluorodecanoic Acid	335-76-2	1.1 J	0.77	1
14473	Pfdoda-Perfluorododecanoic	307-55-1	N.D.	0.43	1
14473	Pfds-Perfluorodecanesulfonate	335-77-3	N.D.	0.52	1
14473	Pfhpa-Perfluoroheptanoic Acid	375-85-9	6.9	0.34	1
14473	Pfhps-Perfluoroheptanesulfonat	375-92-8	N.D.	0.34	1
14473	Pfhxa-Perfluorohexanoic Acid	307-24-4	69	0.34	1
14473	Pfhxs-Perfluorohexanesulfonate	355-46-4	16	0.34	1
14473	Pfna-Perfluorononanoic Acid	375-95-1	1.2 J	0.34	1
14473	Pfoa-Perfluorooctanoic Acid	335-67-1	20	0.26	1
14473	Pfosa-Perfluorooctanesulfonami	754-91-6	N.D.	0.43	1
14473	Pfos-Perfluorooctanesulfonate	1763-23-1	6.6	0.34	1
14473	Pfpea-Perfluoropentanoic Acid	2706-90-3	24	1.7	1
14473	Pfpes-Perfluoropentanesulfonat	2706-91-4	9.6	0.34	1
14473	Pfteda-Perfluorotetradecanoic	376-06-7	N.D.	0.26	1
14473	Pftrda-Perfluorotridecanoic Ac	72629-94-8	N.D.	0.34	1
14473	Pfunda-Perfluoroundecanoic Aci	2058-94-8	N.D.	0.34	1

The recovery for labeled compound used as extraction standards is outside of QC acceptance limits in the method blank associated with this sample as noted on the QC Summary.

The recovery for the labeled compound used as extraction standards is outside the QC acceptance limits as noted on the QC Summary. The following corrective action was taken: The sample was reextracted outside of the required holding time and extraction standards were again outside of the QC acceptance limits. The data is reported from the original extraction. Both sets of data are included in the data package.

Laboratory Sample Analysis Record

CAT No.	Analysis Name	Method	Trial#	Batch#	Analysis Date and Time	Analyst	Dilution Factor
14473	PFAS in Water by LC/MS/MS	EPA 537 Version 1.1 Modified	1	18321001	11/19/2018 11:39	Jason W Knight	1
14091	PFAS Water Prep	EPA 537 Version 1.1 Modified	1	18321001	11/17/2018 08:50	Courtney J Fatta	1

eurofins Lancaster Laboratories Environmental

Analysis Report

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Sample Description: Field Blank (190-17993-3) Water
City of Cadillac WWTP

Project Name: City of Cadillac WWTP

Submittal Date/Time: 11/16/2018 11:40
Collection Date/Time: 11/05/2018 13:10

TestAmerica
ELLE Sample #: WW 9903363
ELLE Group #: 2010210
Matrix: Water

CAT No.	Analysis Name	CAS Number	Result	Method Detection Limit	Dilution Factor
LC/MS/MS Miscellaneous	**EPA 537 Version 1.1 Modified**		**ng/l**	**ng/l**	
14473	4:2 fluorotelomersulfonate	757124-72-4	N.D.	0.89	1
14473	6:2 fluorotelomersulfonate	27619-97-2	N.D.	0.89	1
14473	8:2 fluorotelomersulfonate	39108-34-4	N.D.	1.8	1
14473	NEtFOSAA	2991-50-6	N.D.	0.89	1
	NEtFOSAA is the acronym for N-ethyl perfluorooctanesulfonamidoacetic Acid.				
14473	NMeFOSAA	2355-31-9	N.D.	0.89	1
	NMeFOSAA is the acronym for N-methyl perfluorooctanesulfonamidoacetic Acid.				
14473	Perfluorononanesulfonate(Pfns)	68259-12-1	N.D.	0.53	1
14473	Pfba-Perfluorobutanoic Acid	375-22-4	N.D.	1.8	1
14473	Pfbs-Perfluorobutanesulfonate	375-73-5	N.D.	0.27	1
14473	Pfda-Perfluorodecanoic Acid	335-76-2	N.D.	0.80	1
14473	Pfdoda-Perfluorododecanoic	307-55-1	N.D.	0.44	1
14473	Pfds-Perfluorodecanesulfonate	335-77-3	N.D.	0.53	1
14473	Pfhpa-Perfluoroheptanoic Acid	375-85-9	N.D.	0.35	1
14473	Pfhps-Perfluoroheptanesulfonat	375-92-8	N.D.	0.35	1
14473	Pfhxa-Perfluorohexanoic Acid	307-24-4	N.D.	0.35	1
14473	Pfhxs-Perfluorohexanesulfonate	355-46-4	N.D.	0.35	1
14473	Pfna-Perfluorononanoic Acid	375-95-1	N.D.	0.35	1
14473	Pfoa-Perfluorooctanoic Acid	335-67-1	N.D.	0.27	1
14473	Pfosa-Perfluorooctanesulfonami	754-91-6	N.D.	0.44	1
14473	Pfos-Perfluorooctanesulfonate	1763-23-1	N.D.	0.35	1
14473	Pfpea-Perfluoropentanoic Acid	2706-90-3	N.D.	1.8	1
14473	Pfpes-Perfluoropentanesulfonat	2706-91-4	N.D.	0.35	1
14473	Pfteda-Perfluorotetradecanoic	376-06-7	N.D.	0.27	1
14473	Pftrda-Perfluorotridecanoic Ac	72629-94-8	N.D.	0.35	1
14473	Pfunda-Perfluoroundecanoic Aci	2058-94-8	N.D.	0.35	1

The recovery for labeled compound used as extraction standards is outside of QC acceptance limits in the method blank associated with this sample as noted on the QC Summary.

Laboratory Sample Analysis Record

CAT No.	Analysis Name	Method	Trial#	Batch#	Analysis Date and Time	Analyst	Dilution Factor
14473	PFAS in Water by LC/MS/MS	EPA 537 Version 1.1 Modified	1	18321001	11/19/2018 11:48	Jason W Knight	1
14091	PFAS Water Prep	EPA 537 Version 1.1 Modified	1	18321001	11/17/2018 08:50	Courtney J Fatta	1

Quality Control Summary

Client Name: TestAmerica Group Number: 2010210
Reported: 11/29/2018 16:22

Matrix QC may not be reported if insufficient sample or site-specific QC samples were not submitted. In these situations, to demonstrate precision and accuracy at a batch level, a LCS/LCSD was performed, unless otherwise specified in the method.

All Inorganic Initial Calibration and Continuing Calibration Blanks met acceptable method criteria unless otherwise noted on the Analysis Report.

Method Blank

Analysis Name	Result ng/l	MDL ng/l
Batch number: 18321001	Sample number(s): 9903361-9903363	
4:2 fluorotelomersulfonate	N.D.	1.0
6:2 fluorotelomersulfonate	N.D.	1.0
8:2 fluorotelomersulfonate	N.D.	2.0
NEtFOSAA	N.D.	1.0
NMeFOSAA	N.D.	1.0
Perfluorononanesulfonate(Pfns)	N.D.	0.60
Pfba-Perfluorobutanoic Acid	N.D.	2.0
Pfbs-Perfluorobutanesulfonate	N.D.	0.30
Pfda-Perfluorodecanoic Acid	N.D.	0.90
Pfdoda-Perfluorododecanoic	N.D.	0.50
Pfds-Perfluorodecanesulfonate	N.D.	0.60
Pfhpa-Perfluoroheptanoic Acid	N.D.	0.40
Pfhps-Perfluoroheptanesulfonat	N.D.	0.40
Pfhxa-Perfluorohexanoic Acid	N.D.	0.40
Pfhxs-Perfluorohexanesulfonate	N.D.	0.40
Pfna-Perfluorononanoic Acid	N.D.	0.40
Pfoa-Perfluorooctanoic Acid	N.D.	0.30
Pfosa-Perfluorooctanesulfonami	N.D.	0.50
Pfos-Perfluorooctanesulfonate	N.D.	0.40
Pfpea-Perfluoropentanoic Acid	N.D.	2.0
Pfpes-Perfluoropentanesulfonat	N.D.	0.40
Pfteda-Perfluorotetradecanoic	N.D.	0.30
Pftrda-Perfluorotridecanoic Ac	N.D.	0.40
Pfunda-Perfluoroundecanoic Aci	N.D.	0.40

LCS/LCSD

Analysis Name	LCS Spike Added ng/l	LCS Conc ng/l	LCSD Spike Added ng/l	LCSD Conc ng/l	LCS %REC	LCSD %REC	LCS/LCSD Limits	RPD	RPD Max
Batch number: 18321001	Sample number(s): 9903361-9903363								
4:2 fluorotelomersulfonate	14.94	12.84	14.94	13.35	86	89	82-152	4	30
6:2 fluorotelomersulfonate	15.17	13.88	15.17	13.23	92	87	66-155	5	30
8:2 fluorotelomersulfonate	15.33	14.01	15.33	13.74	91	90	66-148	2	30
NEtFOSAA	5.44	5.43	5.44	4.83	100	89	55-169	12	30
NMeFOSAA	5.44	3.86	5.44	4.27	71	78	62-167	10	30

*- Outside of specification
(1) The result for one or both determinations was less than five times the LOQ.
(2) The unspiked result was more than four times the spike added.

Quality Control Summary

Client Name: TestAmerica
Reported: 11/29/2018 16:22

Group Number: 2010210

LCS/LCSD (continued)

Analysis Name	LCS Spike Added ng/l	LCS Conc ng/l	LCSD Spike Added ng/l	LCSD Conc ng/l	LCS %REC	LCSD %REC	LCS/LCSD Limits	RPD	RPD Max
Perfluorononanesulfonate(Pfns)	5.22	5.01	5.22	5.27	96	101	66-133	5	30
Pfba-Perfluorobutanoic Acid	5.44	5.34	5.44	5.56	98	102	74-142	4	30
Pfbs-Perfluorobutanesulfonate	4.81	4.77	4.81	4.84	99	101	73-128	1	30
Pfda-Perfluorodecanoic Acid	5.44	5.46	5.44	5.76	100	106	69-148	5	30
Pfdoda-Perfluorododecanoic	5.44	6.14	5.44	5.43	113	100	75-136	12	30
Pfds-Perfluorodecanesulfonate	5.24	4.86	5.24	4.99	93	95	60-135	3	30
Pfhpa-Perfluoroheptanoic Acid	5.44	5.38	5.44	5.13	99	94	76-140	5	30
Pfhps-Perfluoroheptanesulfonat	5.18	4.90	5.18	4.97	95	96	64-135	1	30
Pfhxa-Perfluorohexanoic Acid	5.44	5.40	5.44	5.01	99	92	75-135	8	30
Pfhxs-Perfluorohexanesulfonate	5.14	4.91	5.14	5.12	95	100	71-131	4	30
Pfna-Perfluorononanoic Acid	5.44	4.84	5.44	4.94	89	91	72-148	2	30
Pfoa-Perfluorooctanoic Acid	5.44	5.75	5.44	5.58	106	103	72-138	3	30
Pfosa-Perfluorooctanesulfonami	5.44	4.60	5.44	4.65	85	85	65-164	1	30
Pfos-Perfluorooctanesulfonate	5.20	4.54	5.20	5.08	87	98	67-138	11	30
Pfpea-Perfluoropentanoic Acid	5.44	5.39	5.44	4.83	99	89	74-134	11	30
Pfpes-Perfluoropentanesulfonat	5.10	5.12	5.10	5.09	100	100	76-127	1	30
Pfteda-Perfluorotetradecanoic	5.44	5.48	5.44	5.34	101	98	74-135	2	30
Pftrda-Perfluorotridecanoic Ac	5.44	5.68	5.44	5.63	104	104	61-145	1	30
Pfunda-Perfluoroundecanoic Aci	5.44	4.44	5.44	5.34	82	98	75-146	18	30

Labeled Isotope Quality Control

Labeled isotope recoveries which are outside of the QC window are confirmed
unless otherwise noted on the analysis report.

Analysis Name: PFAS in Water by LC/MS/MS
Batch number: 18321001

	13C4-PFBA	13C5-PFPeA	13C3-PFBS	13C2-4:2-FTS	13C5-PFHxA	13C3-PFHxS
9903361	96	61	71	173	74	90
9903362	93	49	55	170	74	90
9903363	94	92	83	99	89	89
Blank	112	113	103	118	110	110
LCS	89	88	78	99	84	79
LCSD	98	100	89	102	94	88
Limits:	33-123	31-157	26-148	21-182	35-138	34-126

	13C4-PFHpA	13C2-6:2-FTS	13C8-PFOA	13C8-PFOS	13C9-PFNA	13C6-PFDA
9903361	98	199*	100	93	124	95
9903362	96	194*	95	90	113	93
9903363	93	133	95	91	99	98
Blank	114	149	113	115	120	111

*- Outside of specification
(1) The result for one or both determinations was less than five times the LOQ.
(2) The unspiked result was more than four times the spike added.

Quality Control Summary

Client Name: TestAmerica
Reported: 11/29/2018 16:22

Group Number: 2010210

Labeled Isotope Quality Control (continued)

Labeled isotope recoveries which are outside of the QC window are confirmed unless otherwise noted on the analysis report.

Analysis Name: PFAS in Water by LC/MS/MS
Batch number: 18321001

	13C4-PFHpA	13C2-6:2-FTS	13C8-PFOA	13C8-PFOS	13C9-PFNA	13C6-PFDA
LCS	89	104	89	90	95	85
LCSD	98	125	99	98	104	97
Limits:	35-126	32-170	48-122	50-121	41-144	47-125

	13C2-8:2-FTS	d3-NMeFOSAA	13C7-PFUnDA	d5-NEtFOSAA	13C2-PFDoDA	13C2-PFTeDA
9903361	130	106	99	107	88	54
9903362	130	113	90	97	77	69
9903363	126	107	98	88	93	100
Blank	126	152*	110	111	104	106
LCS	99	116	89	90	82	80
LCSD	106	108	92	99	92	93
Limits:	27-164	30-127	30-128	30-142	39-130	26-119

	13C8-PFOSA
9903361	53
9903362	50
9903363	104
Blank	111
LCS	87
LCSD	95
Limits:	11-127

*- Outside of specification
(1) The result for one or both determinations was less than five times the LOQ.
(2) The unspiked result was more than four times the spike added.

1042/2010210/9903361-63

TestAmerica Michigan
10448 Citation Drive Suite 200
Brighton, MI 48116
Phone (810) 229-2763 Fax (810) 229-0000

Chain of Custody Record

TestAmerica
THE LEADER IN ENVIRONMENTAL TESTING

Client Information (Sub Contract Lab)	Sampler:	Lab PM: Schafer, Sue	Carrier Tracking No(s):	COC No: 190-20999.1
Client Contact: Shipping/Receiving	Phone:	E-Mail: sue.schafer@testamericainc.com	State of Origin: Michigan	Page: Page 1 of 1
Company: TestAmerica Laboratories, Inc.		Accreditations Required (See note):		Job #: 190-17993-1
Address: 880 Riverside Parkway,	Due Date Requested: 11/29/2018	**Analysis Requested**		Preservation Codes:
City: West Sacramento	TAT Requested (days):			
State, Zip: CA, 95605				
Phone: 916-373-5600(Tel) 916-372-1059(Fax)	PO #:			
Email:	WO #:			
Project Name: City of Cadillac WWTP	Project #: 19001884			
Site:	SSOW#:			Other:

Preservation Codes:
A - HCL, B - NaOH, C - Zn Acetate, D - Nitric Acid, E - NaHSO4, F - MeOH, G - Amchlor, H - Ascorbic Acid, I - Ice, J - DI Water, K - EDTA, L - EDA
M - Hexane, N - None, O - AsNaO2, P - Na2O4S, Q - Na2SO3, R - Na2S2O3, S - H2SO4, T - TSP Dodecahydrate, U - Acetone, V - MCAA, W - pH 4-5, Z - other (specify)

Sample Identification - Client ID (Lab ID)	Sample Date	Sample Time	Sample Type (C=comp, G=grab)	Matrix (W=water, S=solid, O=waste/oil, BT=Tissue, A=Air)	Field Filtered Sample (Yes or No)	Perform MS/MSD (Yes or No)	PFC_IDA/3535_PFC PFAS, Standard List (24 Analytes)	Total Number of containers	Special Instructions/Note:
			Preservation Code:						
Effluent (190-17993-1)	11/5/18	13:15 Eastern		Water			X	2	
Effluent Duplicate (190-17993-2)	11/5/18	13:15 Eastern		Water			X	2	
Field Blank (190-17993-3)	11/5/18	13:10 Eastern		Water			X	2	

Note: Since laboratory accreditations are subject to change, TestAmerica Laboratories, Inc. places the ownership of method, analyte & accreditation compliance upon out subcontract laboratories. This sample shipment is forwarded under chain-of-custody. If the laboratory does not currently maintain accreditation in the State of Origin listed above for analysis/tests/matrix being analyzed, the samples must be shipped back to the TestAmerica laboratory or other instructions will be provided. Any changes to accreditation status should be brought to TestAmerica Laboratories, Inc. attention immediately. If all requested accreditations are current to date, return the signed Chain of Custody attesting to said compliance to TestAmerica Laboratories, Inc.

Possible Hazard Identification
Unconfirmed

Sample Disposal (A fee may be assessed if samples are retained longer than 1 month)
☐ *Return To Client* ☐ *Disposal By Lab* ☐ *Archive For ____ Months*

Deliverable Requested: I, II, III, IV, Other (specify) Primary Deliverable Rank: 2

Special Instructions/QC Requirements:

Empty Kit Relinquished by:	Date:	Time:		Method of Shipment:	
Relinquished by: [signature]	Date/Time: 11/6/18 1420	Company: TAL	Received by: [signature]	Date/Time: 11-7-18 1005	Company: TA Sac
Relinquished by: [signature]	Date/Time: 11/15/18 1630	Company: TA Sac	Received by:	Date/Time:	Company:
Relinquished by:	Date/Time:	Company:	Received by: [signature]	Date/Time: 11/16/18 1140	Company:
Custody Seals Intact: Δ Yes Δ No	Custody Seal No.:		Cooler Temperature(s) °C and Other Remarks: 3.1		

Ver: 09/20/2016
12/10/2018

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Sample Administration Receipt Documentation Log

Doc Log ID: 233532

Group Number(s): 2010210

Client: Test America

Delivery and Receipt Information

Delivery Method: Fed Ex

Arrival Timestamp: 11/16/2018 11:40

Number of Packages: 2

Number of Projects: 9

Arrival Condition Summary

Shipping Container Sealed:	Yes
Custody Seal Present:	Yes
Custody Seal Intact:	Yes
Samples Chilled:	Yes
Paperwork Enclosed:	Yes
Samples Intact:	Yes
Missing Samples:	No
Extra Samples:	No
Discrepancy in Container Qty on COC:	No

Sample IDs on COC match Containers:	Yes
Sample Date/Times match COC:	Yes
VOA Vial Headspace ≥ 6mm:	N/A
Total Trip Blank Qty:	0
Air Quality Samples Present:	No

Unpacked by Ariel Garcia (15332) at 13:22 on 11/16/2018

Samples Chilled Details

Thermometer Types: *DT = Digital (Temp. Bottle)* *IR = Infrared (Surface Temp)* *All Temperatures in °C.*

Cooler #	Thermometer ID	Corrected Temp	Therm. Type	Ice Type	Ice Present?	Ice Container	Elevated Temp?
1	DT42-03	0.5	DT	Wet	Y	Loose	N
2	DT42-03	0.6	DT	Wet	Y	Loose	N

2425 New Holland Pike
Lancaster, PA 17605-2425

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www.LancasterLabs.com

12/10/2018

Explanation of Symbols and Abbreviations

The following defines common symbols and abbreviations used in reporting technical data:

BMQL	Below Minimum Quantitation Level	**mL**	milliliter(s)
C	degrees Celsius	**MPN**	Most Probable Number
cfu	colony forming units	**N.D.**	non-detect
CP Units	cobalt-chloroplatinate units	**ng**	nanogram(s)
F	degrees Fahrenheit	**NTU**	nephelometric turbidity units
g	gram(s)	**pg/L**	picogram/liter
IU	International Units	**RL**	Reporting Limit
kg	kilogram(s)	**TNTC**	Too Numerous To Count
L	liter(s)	**µg**	microgram(s)
lb.	pound(s)	**µL**	microliter(s)
m3	cubic meter(s)	**umhos/cm**	micromhos/cm
meq	milliequivalents	**MCL**	Maximum Contamination Limit
mg	milligram(s)		

<	less than
>	greater than
ppm	parts per million - One ppm is equivalent to one milligram per kilogram (mg/kg) or one gram per million grams. For aqueous liquids, ppm is usually taken to be equivalent to milligrams per liter (mg/l), because one liter of water has a weight very close to a kilogram. For gases or vapors, one ppm is equivalent to one microliter per liter of gas.
ppb	parts per billion
Dry weight basis	Results printed under this heading have been adjusted for moisture content. This increases the analyte weight concentration to approximate the value present in a similar sample without moisture. All other results are reported on an as-received basis.

Analytical test results meet all requirements of the associated regulatory program (i.e., NELAC (TNI), DoD, and ISO 17025) unless otherwise noted under the individual analysis.

Measurement uncertainty values, as applicable, are available upon request.

Tests results relate only to the sample tested. Clients should be aware that a critical step in a chemical or microbiological analysis is the collection of the sample. Unless the sample analyzed is truly representative of the bulk of material involved, the test results will be meaningless. If you have questions regarding the proper techniques of collecting samples, please contact us. We cannot be held responsible for sample integrity, however, unless sampling has been performed by a member of our staff.

This report shall not be reproduced except in full, without the written approval of the laboratory.

Times are local to the area of activity. Parameters listed in the 40 CFR Part 136 Table II as “analyze immediately” are not performed within 15 minutes.

WARRANTY AND LIMITS OF LIABILITY - In accepting analytical work, we warrant the accuracy of test results for the sample as submitted. THE FOREGOING EXPRESS WARRANTY IS EXCLUSIVE AND IS GIVEN IN LIEU OF ALL OTHER WARRANTIES, EXPRESSED OR IMPLIED. WE DISCLAIM ANY OTHER WARRANTIES, EXPRESSED OR IMPLIED, INCLUDING A WARRANTY OF FITNESS FOR PARTICULAR PURPOSE AND WARRANTY OF MERCHANTABILITY. IN NO EVENT SHALL EUROFINS LANCASTER LABORATORIES ENVIRONMENTAL, LLC BE LIABLE FOR INDIRECT, SPECIAL, CONSEQUENTIAL, OR INCIDENTAL DAMAGES INCLUDING, BUT NOT LIMITED TO, DAMAGES FOR LOSS OF PROFIT OR GOODWILL REGARDLESS OF (A) THE NEGLIGENCE (EITHER SOLE OR CONCURRENT) OF EUROFINS LANCASTER LABORATORIES ENVIRONMENTAL AND (B) WHETHER EUROFINS LANCASTER LABORATORIES ENVIRONMENTAL HAS BEEN INFORMED OF THE POSSIBILITY OF SUCH DAMAGES. We accept no legal responsibility for the purposes for which the client uses the test results. No purchase order or other order for work shall be accepted by Eurofins Lancaster Laboratories Environmental which includes any conditions that vary from the Standard Terms and Conditions, and Eurofins Lancaster Laboratories Environmental hereby objects to any conflicting terms contained in any acceptance or order submitted by client.

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Data Qualifiers

Qualifier	Definition
C	Result confirmed by reanalysis
D1	Indicates for dual column analyses that the result is reported from column 1
D2	Indicates for dual column analyses that the result is reported from column 2
E	Concentration exceeds the calibration range
K1	Initial Calibration Blank is above the QC limit and the sample result is ND
K2	Continuing Calibration Blank is above the QC limit and the sample result is ND
K3	Initial Calibration Verification is above the QC limit and the sample result is ND
K4	Continuing Calibration Verification is above the QC limit and the sample result is ND
J (or G, I, X)	Estimated value >= the Method Detection Limit (MDL or DL) and < the Limit of Quantitation (LOQ or RL)
P	Concentration difference between the primary and confirmation column >40%. The lower result is reported.
P^	Concentration difference between the primary and confirmation column > 40%. The higher result is reported.
U	Analyte was not detected at the value indicated
V	Concentration difference between the primary and confirmation column >100%. The reporting limit is raised due to this disparity and evident interference.
W	The dissolved oxygen uptake for the unseeded blank is greater than 0.20 mg/L.
Z	Laboratory Defined - see analysis report

Additional Organic and Inorganic CLP qualifiers may be used with Form 1 reports as defined by the CLP methods.
Qualifiers specific to Dioxin/Furans and PCB Congeners are detailed on the individual Analysis Report.

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Sample Administration Receipt Documentation Log

Doc Log ID: 233532

Group Number(s): 2010210

Client: Test America

Delivery and Receipt Information

Delivery Method: Fed Ex

Arrival Timestamp: 11/16/2018 11:40

Number of Packages: 2

Number of Projects: 9

Arrival Condition Summary

Shipping Container Sealed:	Yes
Custody Seal Present:	Yes
Custody Seal Intact:	Yes
Samples Chilled:	Yes
Paperwork Enclosed:	Yes
Samples Intact:	Yes
Missing Samples:	No
Extra Samples:	No
Discrepancy in Container Qty on COC:	No

Sample IDs on COC match Containers:	Yes
Sample Date/Times match COC:	Yes
VOA Vial Headspace ≥ 6mm:	N/A
Total Trip Blank Qty:	0
Air Quality Samples Present:	No

Unpacked by Ariel Garcia (15332) at 13:22 on 11/16/2018

Samples Chilled Details

Thermometer Types: *DT = Digital (Temp. Bottle)* *IR = Infrared (Surface Temp)* *All Temperatures in °C.*

Cooler #	Thermometer ID	Corrected Temp	Therm. Type	Ice Type	Ice Present?	Ice Container	Elevated Temp?
1	DT42-03	0.5	DT	Wet	Y	Loose	N
2	DT42-03	0.6	DT	Wet	Y	Loose	N

2425 New Holland Pike
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Glossary

Abbreviation	These commonly used abbreviations may or may not be present in this report.
¤	Listed under the "D" column to designate that the result is reported on a dry weight basis
%R	Percent Recovery
CFL	Contains Free Liquid
CNF	Contains No Free Liquid
DER	Duplicate Error Ratio (normalized absolute difference)
Dil Fac	Dilution Factor
DL	Detection Limit (DoD/DOE)
DL, RA, RE, IN	Indicates a Dilution, Re-analysis, Re-extraction, or additional Initial metals/anion analysis of the sample
DLC	Decision Level Concentration (Radiochemistry)
EDL	Estimated Detection Limit (Dioxin)
LOD	Limit of Detection (DoD/DOE)
LOQ	Limit of Quantitation (DoD/DOE)
MDA	Minimum Detectable Activity (Radiochemistry)
MDC	Minimum Detectable Concentration (Radiochemistry)
MDL	Method Detection Limit
ML	Minimum Level (Dioxin)
NC	Not Calculated
ND	Not Detected at the reporting limit (or MDL or EDL if shown)
PQL	Practical Quantitation Limit
QC	Quality Control
RER	Relative Error Ratio (Radiochemistry)
RL	Reporting Limit or Requested Limit (Radiochemistry)
RPD	Relative Percent Difference, a measure of the relative difference between two points
TEF	Toxicity Equivalent Factor (Dioxin)
TEQ	Toxicity Equivalent Quotient (Dioxin)

TestAmerica Michigan
10448 Citation Drive Suite 200
Brighton, MI 48116
Phone (810) 229-2763 Fax (810) 229-0000

Chain of Custody Record

TestAmerica
THE LEADER IN ENVIRONMENTAL TESTING

Client Information
Client Contact: Cindy Tomaszewski
Company: City of Cadillac Utilities
Address: 1121 Plett Road
City: Cadillac
State, Zip: MI, 49601
Phone: 231-775-2368
Email: lab@cadillac-mi.net
Project Name: City of Cadillac
Site: WWTP

Sampler: CATomaszewski
Phone: 231-775-2368
Lab PM: Schafer, Sue
E-Mail: sue.schafer@testamericainc.com
Carrier Tracking No(s):
COC No: 190-21716-912.1
Page: Page 1 of 1
Job #:

Due Date Requested:
TAT Requested (days):
PO #: ~~2019-018~~ TBD
WO #:
Project #: 19001884
SSOW#:

Analysis Requested

Sample Identification	Sample Date	Sample Time	Sample Type (C=comp, G=grab)	Matrix (W=water, S=solid, O=waste/oil, BT=Tissue, A=Air)	Field Filtered Sample (Yes or No)	Perform MS/MSD (Yes or No)	PFC_IDA - PFAS, Standard List (24 Analytes)	Total Number of containers	Special Instructions/Note:
			Preservation Code:				N		
EFFLUENT	11-5-18		G	Water	N	N	X	2	
EFFLUENT DUPLIC.	11-5-18		G	Water	N	N	X	2	
FIELD BLANK	11-5-18		G	Water	N	N	X	2	
				~~Water~~					

Preservation Codes:
A - HCL
B - NaOH
C - Zn Acetate
D - Nitric Acid
E - NaHSO4
F - MeOH
G - Amchlor
H - Ascorbic Acid
I - Ice
J - DI Water
K - EDTA
L - EDA
M - Hexane
N - None
O - AsNaO2
P - Na2O4S
Q - Na2SO3
R - Na2S2O3
S - H2SO4
T - TSP Dodecahydrate
U - Acetone
V - MCAA
W - pH 4-5
Z - other (specify)
Other:

190-17993 Chain of Custody

Possible Hazard Identification
☐ *Non-Hazard* ☐ *Flammable* ☐ *Skin Irritant* ☐ *Poison B* ☐ *Unknown* ☐ *Radiological*

Sample Disposal (A fee may be assessed if samples are retained longer than 1 month)
☐ *Return To Client* ☒ *Disposal By Lab* ☐ *Archive For* ______ *Months*

Deliverable Requested: I, II, III, IV, Other (specify)
Special Instructions/QC Requirements:

Empty Kit Relinquished by:	Date:		Time:	Method of Shipment: UPS Ground	
Relinquished by: CATomaszewski	Date/Time: 11-5-18/1400	Company: City of Cadillac	Received by: [signature]	Date/Time: 11/6/18 9:35	Company: TAL MI
Relinquished by:	Date/Time:	Company	Received by:	Date/Time:	Company
Relinquished by:	Date/Time:	Company	Received by:	Date/Time:	Company

Custody Seals Intact: Δ Yes Δ No
Custody Seal No.:
Cooler Temperature(s) °C and Other Remarks:

Ver: 08/04/2016

TestAmerica

Cooler/Sample Receipt

(AFTER HOURS receipt: complete gray areas. Place cooler in walk-in; place this form in Receiving in-box. Date/Time rec'd: ____ Initials: ____)

❑ MSDS or Known Hazard Information Supplied by Client
❑ Bottle stickers applied ❑ ELEMENT comment entered ❑ MSDS/COC scanned/emailed to EHSS

❑ Discrepancies — Client ID: City of Cadillac

❑ Short Hold — Work Order #: 190-17993

❑ Rush ❑24hr ❑2day ❑3day ❑5day ❑other____

Receipt evaluation performed by - Initials: AMY Date: 11/6/18 Time: 9:35

Method of Shipment:
❑ Walk-In Client ❑ TestAmerica Field/Courier
❑ Other Client/3rd Party Courier ____
❑ Fed Ex Tracking # ____
☒ UPS Tracking # 1Z E47 491 03 5387 7109
❑ Other Ground

Shipping Container Type:
☒ Cooler ❑ Box
❑ None ❑ Other____

Packing Materials:
☒ Plastic Bags ❑ Foam
❑ Bubble Wrap ❑ Paper
❑ Packing Peanuts ❑ None
☒ Other Air Packs

Custody Seals Intact:
☒ Yes 615588/885 ❑ No
❑ N/A (not used or required)

Cooling Materials:
☒ Ice (solid) ❑ Ice (Melted)
❑ Blue Ice ❑ None
❑ Other____

Bacteriological Samples	Temp (oC) Corrected	Frozen	Received within 2 hours	Sample Flagged
		yes no	yes no	yes no

Receipt Temperatures ❑ Check if Additional Sheets Required

C.F.

Thermometer ID	Observed (°C)	Corrected (°C)	Temp Blank	Sample Temp	Received on same day sampled?	Acceptable?*	Cooler ID	Note Affected Samples if temperature not acceptable
140252483			❑	❑	❑Yes ❑No	❑Yes ❑No		
140252476			❑	❑	❑Yes ❑No	❑Yes ❑No		
-0.7 CP313207	3.8	3.1	❑	☒	❑Yes ❑No	☒Yes ❑No		

* Receipt temperatures are considered acceptable if the samples are received on the same day they were collected & show signs that the cooling process has started. Temperature acceptance for most tests is ≤6.0°C, but not frozen. For additional information, please refer to SOP DT-SCA-004 *Sample Receipt and Login, Attachment 2 – Holding Times, Preservation and Container Requirements*

Receipt Questions**	Y	N	n/a	"No" answers require additional comment
COC present & TA receipt signature, date, & time properly documented?	✓			3 samples rec'd; cross offs not initialed.
Containers & labels in good condition? *(unbroken, not leaking, appropriately filled, labels legible & attached)*	✓			
Appropriate containers used & adequate volume provided?	✓			Preserved Bottles Checked with pH Strips* Yes No
Number of sample containers match COC?	✓			
Samples received within hold time?	✓			
Samples submitted for GRO and Volatiles analyses (8260, 624, 524) received without headspace?			✓	
Was a Trip Blank received with VOA samples?			✓	
Were the samples free of any questionable physical conformities? For example, field duplicates or multiple bottles of the same sample do not significantly vary in appearance (color, proportion of solids, etc.)	✓			
Were the COC, bottle labels, and all other items free of all other discrepancies or issues that would need to be addressed with the Project Manager and/or Client?	✓	**		** No Times on COC ... Times from bottles Eff 13:15 Dup 13:15 FB 13:10

** May not be applicable if samples are not for compliance testing — * Excludes FOG, Volatiles, TOC Vials

Client Contact Record

Contact via: ❑ Phone ❑ Email ❑ Other____ Person Contacted: ____ Date/Time: ____

❑ Discrepancy allowance agreement is on record in the client project file

Discussion/Resolution:

Any additional documentation and clarification from client must be noted in the narrative and/or scanned into the COC directory.

Reviewed by PM Signature [signature] Date 11/6/18

WI Page 1 of 1

WI No. DT-SCA-WI-001.13 effective 06/11/12

TestAmerica Michigan
10448 Citation Drive Suite 200
Brighton, MI 48116
Phone (810) 229-2763 Fax (810) 229-0000

Chain of Custody Record

TestAmerica
THE LEADER IN ENVIRONMENTAL TESTING

Client Information (Sub Contract Lab)	Sampler:	Lab PM: Schafer, Sue	Carrier Tracking No(s):	COC No: 190-20999.1
Client Contact: Shipping/Receiving	Phone:	E-Mail: sue.schafer@testamericainc.com	State of Origin: Michigan	Page: Page 1 of 1
Company: TestAmerica Laboratories, Inc.		Accreditations Required (See note):		Job #: 190-17993-1
Address: 880 Riverside Parkway, ,	Due Date Requested: 11/29/2018	**Analysis Requested**		**Preservation Codes:**
City: West Sacramento	TAT Requested (days):			A - HCL, B - NaOH, C - Zn Acetate, D - Nitric Acid, E - NaHSO4, F - MeOH, G - Amchlor, H - Ascorbic Acid, I - Ice, J - DI Water, K - EDTA, L - EDA; M - Hexane, N - None, O - AsNaO2, P - Na2O4S, Q - Na2SO3, R - Na2S2O3, S - H2SO4, T - TSP Dodecahydrate, U - Acetone, V - MCAA, W - pH 4-5, Z - other (specify)
State, Zip: CA, 95605				
Phone: 916-373-5600(Tel) 916-372-1059(Fax)	PO #:			
Email:	WO #:			
Project Name: City of Cadillac WWTP	Project #: 19001884			
Site:	SSOW#:			Other:

Sample Identification - Client ID (Lab ID)	Sample Date	Sample Time	Sample Type (C=comp, G=grab)	Matrix (W=water, S=solid, O=waste/oil, BT=Tissue, A=Air)	Field Filtered Sample (Yes or No)	Perform MS/MSD (Yes or No)	PFC_IDA/3535_PFC PFAS, Standard List (24 Analytes)	Total Number of containers	Special Instructions/Note:
			Preservation Code:						
Effluent (190-17993-1)	11/5/18	13:15 Eastern		Water			X	2	
Effluent Duplicate (190-17993-2)	11/5/18	13:15 Eastern		Water			X	2	
Field Blank (190-17993-3)	11/5/18	13:10 Eastern		Water			X	2	

Note: Since laboratory accreditations are subject to change, TestAmerica Laboratories, Inc. places the ownership of method, analyte & accreditation compliance upon out subcontract laboratories. This sample shipment is forwarded under chain-of-custody. If the laboratory does not currently maintain accreditation in the State of Origin listed above for analysis/tests/matrix being analyzed, the samples must be shipped back to the TestAmerica laboratory or other instructions will be provided. Any changes to accreditation status should be brought to TestAmerica Laboratories, Inc. attention immediately. If all requested accreditations are current to date, return the signed Chain of Custody attesting to said complicance to TestAmerica Laboratories, Inc.

Possible Hazard Identification Unconfirmed	*Sample Disposal (A fee may be assessed if samples are retained longer than 1 month)* ☐ Return To Client ☐ Disposal By Lab ☐ Archive For ______ Months
Deliverable Requested: I, II, III, IV, Other (specify) — Primary Deliverable Rank: 2	Special Instructions/QC Requirements:

Empty Kit Relinquished by:	Date:		Time:	Method of Shipment:	
Relinquished by: [signature]	Date/Time: 11/6/18 14:30	Company: TAE	Received by: [signature]	Date/Time: 11-7-18 1005	Company: TA Sac
Relinquished by:	Date/Time:	Company	Received by:	Date/Time:	Company
Relinquished by:	Date/Time:	Company	Received by:	Date/Time:	Company
Custody Seals Intact: Δ Yes Δ No	Custody Seal No.:		Cooler Temperature(s) °C and Other Remarks: 3.1		

Ver: 09/20/2016